



1. Description

1.1. Project

Project Name	SensnornodeV1
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	06/16/2026

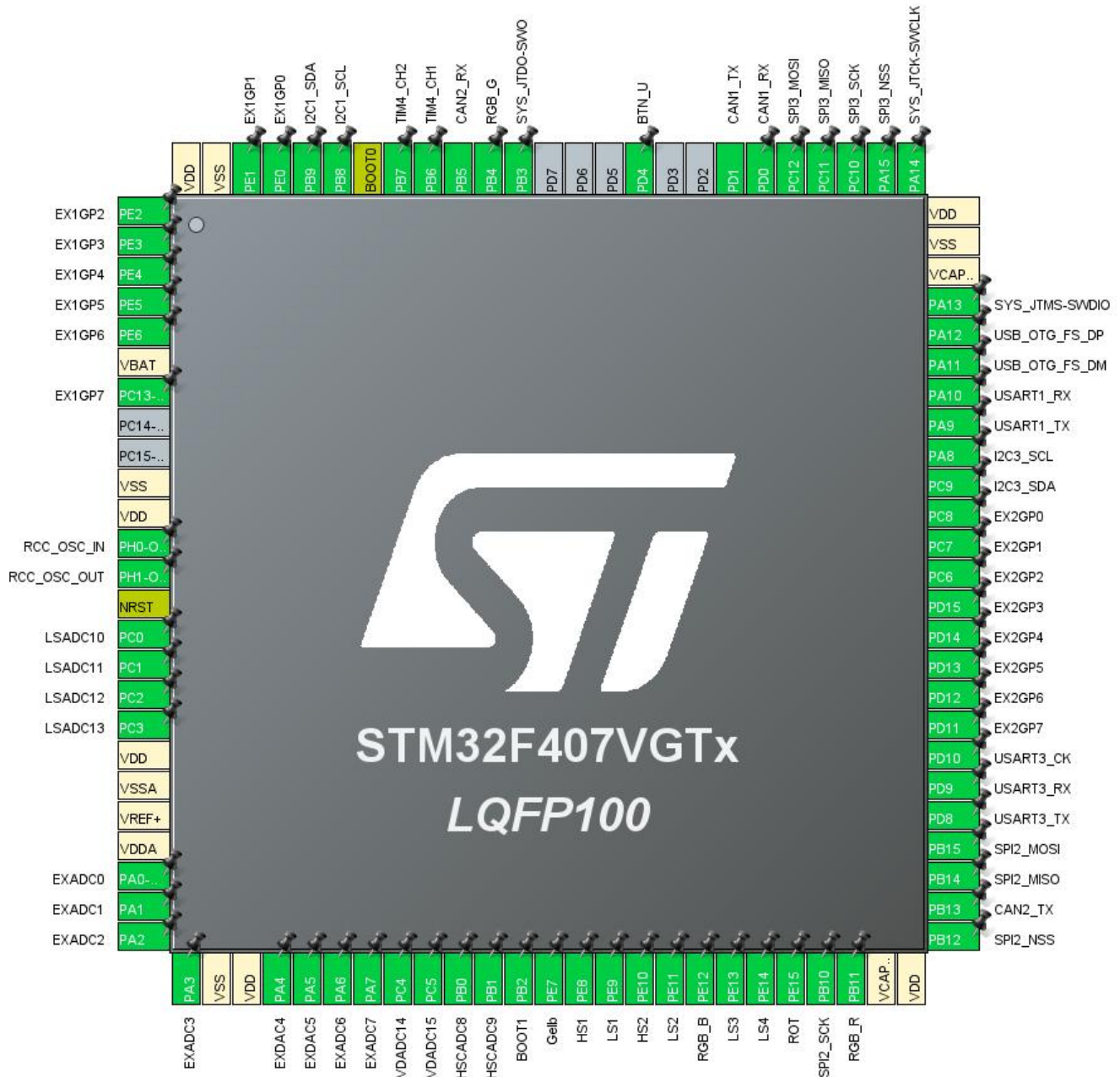
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F407/417
MCU name	STM32F407VGTx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	Arm Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2 *	I/O	GPIO_Output	EX1GP2
2	PE3 *	I/O	GPIO_Output	EX1GP3
3	PE4 *	I/O	GPIO_Output	EX1GP4
4	PE5 *	I/O	GPIO_Output	EX1GP5
5	PE6 *	I/O	GPIO_Output	EX1GP6
6	VBAT	Power		
7	PC13-ANTI_TAMP *	I/O	GPIO_Output	EX1GP7
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
14	NRST	Reset		
15	PC0	I/O	ADC1_IN10	LSADC10
16	PC1	I/O	ADC1_IN11	LSADC11
17	PC2	I/O	ADC1_IN12	LSADC12
18	PC3	I/O	ADC1_IN13	LSADC13
19	VDD	Power		
20	VSSA	Power		
21	VREF+	Power		
22	VDDA	Power		
23	PA0-WKUP	I/O	ADC1_IN0	EXADC0
24	PA1	I/O	ADC1_IN1	EXADC1
25	PA2	I/O	ADC1_IN2	EXADC2
26	PA3	I/O	ADC1_IN3	EXADC3
27	VSS	Power		
28	VDD	Power		
29	PA4	I/O	DAC_OUT1	EXDAC4
30	PA5	I/O	DAC_OUT2	EXDAC5
31	PA6	I/O	ADC1_IN6	EXADC6
32	PA7	I/O	ADC1_IN7	EXADC7
33	PC4	I/O	ADC1_IN14	VDADC14
34	PC5	I/O	ADC1_IN15	VDADC15
35	PB0	I/O	ADC1_IN8	HSCADC8
36	PB1	I/O	ADC1_IN9	HSCADC9
37	PB2 *	I/O	GPIO_Input	BOOT1
38	PE7 *	I/O	GPIO_Output	Gelb

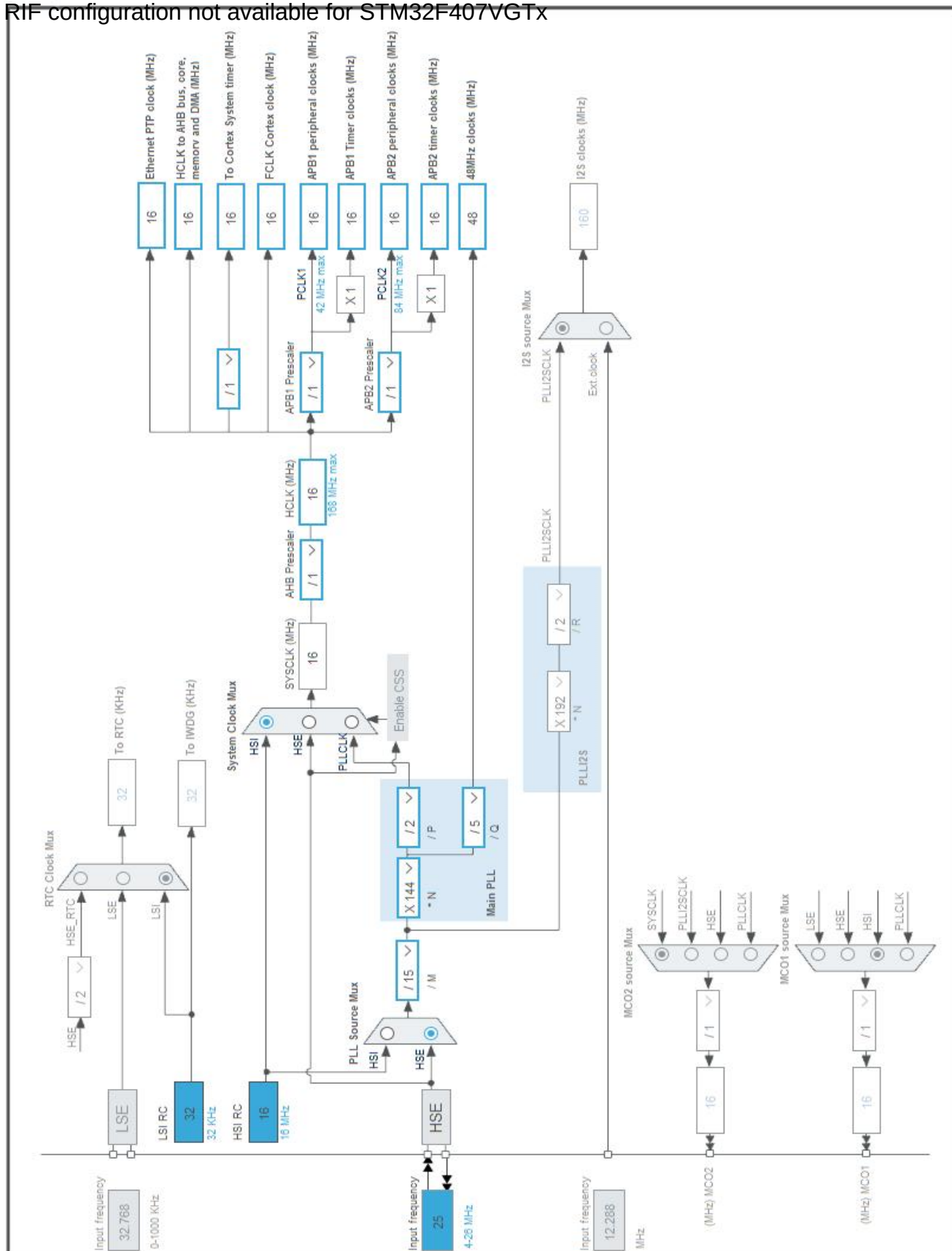
Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
39	PE8	I/O	TIM1_CH1N	HS1
40	PE9	I/O	TIM1_CH1	LS1
41	PE10	I/O	TIM1_CH2N	HS2
42	PE11	I/O	TIM1_CH2	LS2
43	PE12	I/O	TIM1_CH3N	RGB_B
44	PE13	I/O	TIM1_CH3	LS3
45	PE14	I/O	TIM1_CH4	LS4
46	PE15 *	I/O	GPIO_Output	ROT
47	PB10	I/O	SPI2_SCK	
48	PB11	I/O	TIM2_CH4	RGB_R
49	VCAP_1	Power		
50	VDD	Power		
51	PB12	I/O	SPI2_NSS	
52	PB13	I/O	CAN2_TX	
53	PB14	I/O	SPI2_MISO	
54	PB15	I/O	SPI2_MOSI	
55	PD8	I/O	USART3_TX	
56	PD9	I/O	USART3_RX	
57	PD10	I/O	USART3_CK	
58	PD11 *	I/O	GPIO_Output	EX2GP7
59	PD12 *	I/O	GPIO_Output	EX2GP6
60	PD13 *	I/O	GPIO_Output	EX2GP5
61	PD14 *	I/O	GPIO_Output	EX2GP4
62	PD15 *	I/O	GPIO_Output	EX2GP3
63	PC6 *	I/O	GPIO_Output	EX2GP2
64	PC7 *	I/O	GPIO_Output	EX2GP1
65	PC8 *	I/O	GPIO_Output	EX2GP0
66	PC9	I/O	I2C3_SDA	
67	PA8	I/O	I2C3_SCL	
68	PA9	I/O	USART1_TX	
69	PA10	I/O	USART1_RX	
70	PA11	I/O	USB_OTG_FS_DM	
71	PA12	I/O	USB_OTG_FS_DP	
72	PA13	I/O	SYS_JTMS-SWDIO	
73	VCAP_2	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14	I/O	SYS_JTCK-SWCLK	
77	PA15	I/O	SPI3_NSS	

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
78	PC10	I/O	SPI3_SCK	
79	PC11	I/O	SPI3_MISO	
80	PC12	I/O	SPI3_MOSI	
81	PD0	I/O	CAN1_RX	
82	PD1	I/O	CAN1_TX	
85	PD4 *	I/O	GPIO_Input	BTN_U
89	PB3	I/O	SYS_JTDO-SWO	
90	PB4	I/O	TIM3_CH1	RGB_G
91	PB5	I/O	CAN2_RX	
92	PB6	I/O	TIM4_CH1	
93	PB7	I/O	TIM4_CH2	
94	BOOT0	Boot		
95	PB8	I/O	I2C1_SCL	
96	PB9	I/O	I2C1_SDA	
97	PE0 *	I/O	GPIO_Output	EX1GP0
98	PE1 *	I/O	GPIO_Output	EX1GP1
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32F407VGTx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F407/417
MCU	STM32F407VGTx
Datasheet	DS8626 Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

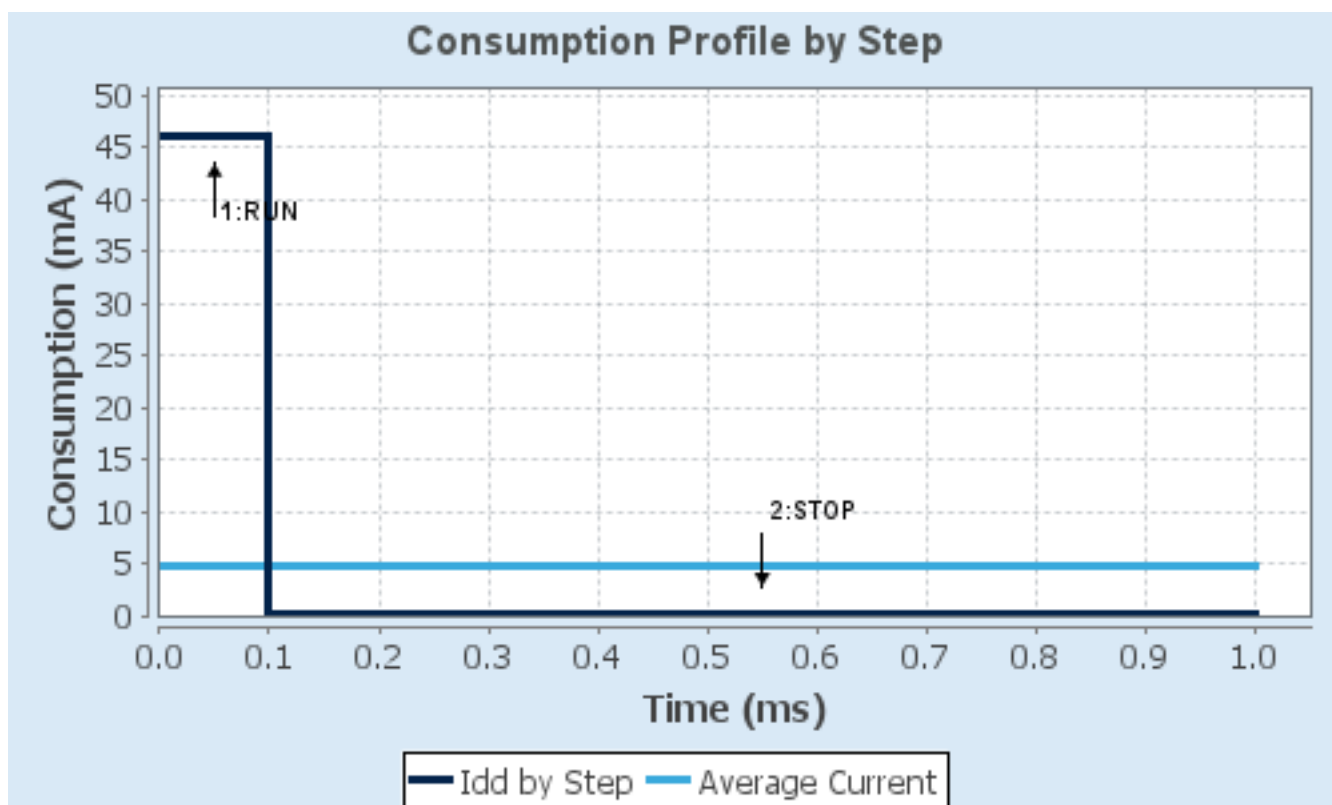
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	FLASH	n/a
CPU Frequency	168 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	46 mA	280 μ A
Duration	0.1 ms	0.9 ms
DMIPS	210.0	0.0
Ta Max	98.47	104.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.85 mA
Battery Life	29 days, 4 hours	Average DMIPS	210.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	SensnornodeV1
Project Folder	C:\Users\livenh\Programme\Coding\STM32\Code
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_CAN1_Init	CAN1
5	MX_CAN2_Init	CAN2
6	MX_DAC_Init	DAC
7	MX_I2C1_Init	I2C1
8	MX_I2C3_Init	I2C3
9	MX_SPI2_Init	SPI2
10	MX_SPI3_Init	SPI3
11	MX_TIM1_Init	TIM1

Rank	Function Name	Peripheral Instance Name
12	MX_TIM2_Init	TIM2
13	MX_TIM3_Init	TIM3
14	MX_TIM4_Init	TIM4
15	MX_USART1_UART_Init	USART1
16	MX_USART3_Init	USART3
17	MX_USB_OTG_FS_PCD_Init	USB_OTG_FS

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN0

mode: IN1

mode: IN2

mode: IN3

mode: IN6

mode: IN7

mode: IN8

mode: IN9

mode: IN10

mode: IN11

mode: IN12

mode: IN13

mode: IN14

mode: IN15

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode
ADC_Settings:	
Clock Prescaler	PCLK2 divided by 2
Resolution	12 bits (15 ADC Clock cycles)
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	EOC flag at the end of single channel conversion

ADC_Regular_ConversionMode:

Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 10 *
Sampling Time	3 Cycles

ADC_Injected_ConversionMode:

Number Of Conversions	0
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WatchDog:

Enable Analog WatchDog Mode false

3.2. CAN1

mode: Activated

3.2.1. Parameter Settings:

Bit Timings Parameters:

Prescaler (for Time Quantum)	16
Time Quantum	1000.0 *
Time Quanta in Bit Segment 1	1 Time
Time Quanta in Bit Segment 2	1 Time
Time for one Bit	3000 *
Baud Rate	333333 *
ReSynchronization Jump Width	1 Time

Basic Parameters:

Time Triggered Communication Mode	Disable
Automatic Bus-Off Management	Disable
Automatic Wake-Up Mode	Disable
Automatic Retransmission	Disable
Receive Fifo Locked Mode	Disable
Transmit Fifo Priority	Disable

Advanced Parameters:

Operating Mode	Normal
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3.3. CAN2

mode: Activated

3.3.1. Parameter Settings:

Bit Timings Parameters:

Prescaler (for Time Quantum)	16
Time Quantum	1000.0 *
Time Quanta in Bit Segment 1	1 Time
Time Quanta in Bit Segment 2	1 Time
Time for one Bit	3000 *
Baud Rate	333333 *

ReSynchronization Jump Width 1 Time

Basic Parameters:

Time Triggered Communication Mode	Disable
Automatic Bus-Off Management	Disable
Automatic Wake-Up Mode	Disable
Automatic Retransmission	Disable
Receive Fifo Locked Mode	Disable
Transmit Fifo Priority	Disable

Advanced Parameters:

Operating Mode	Normal
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3.4. DAC

mode: OUT1 Configuration

mode: OUT2 Configuration

3.4.1. Parameter Settings:

DAC Out1 Settings:

Output Buffer	Enable
Trigger	None

DAC Out2 Settings:

Output Buffer	Enable
Trigger	None

3.5. I2C1

I2C: I2C

3.5.1. Parameter Settings:

Master Features:

I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

3.6. I2C3

I2C: I2C

3.6.1. Parameter Settings:

Master Features:

I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

3.7. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.7.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	0 WS (1 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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3.8. SPI2

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Input Signal

3.8.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	8.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Input Hardware

3.9. SPI3

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Input Signal

3.9.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	8.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Input Hardware

3.10. SYS

Debug: Trace Asynchronous Sw
Timebase Source: SysTick

3.11. TIM1

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Channel4: PWM Generation CH4

3.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off
Dead Time	0

PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
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Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.12. TIM2

Channel4: Forced Output CH4

3.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Forced Output Channel 4:

Mode	Forced Active
Pulse (32 bits value)	0

Output compare preload	Disable
CH Polarity	High

3.13. TIM3

Channel1: PWM Generation CH1

3.13.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.14. TIM4

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

3.14.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.15. USART1

Mode: Asynchronous

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

3.16. USART3

Mode: Synchronous

3.16.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
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Clock Parameters:

Clock Polarity	Low
Clock Phase	One Edge
Clock Last Bit	Disable

3.17. USB_OTG_FS

Mode: Device_Only

3.17.1. Parameter Settings:

Speed	Device Full Speed 12MBit/s
Low power	Disabled
Link Power Management	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_IN10	Analog mode	No pull-up and no pull-down	n/a	LSADC10
	PC1	ADC1_IN11	Analog mode	No pull-up and no pull-down	n/a	LSADC11
	PC2	ADC1_IN12	Analog mode	No pull-up and no pull-down	n/a	LSADC12
	PC3	ADC1_IN13	Analog mode	No pull-up and no pull-down	n/a	LSADC13
	PA0-WKUP	ADC1_IN0	Analog mode	No pull-up and no pull-down	n/a	EXADC0
	PA1	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	EXADC1
	PA2	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	EXADC2
	PA3	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	EXADC3
	PA6	ADC1_IN6	Analog mode	No pull-up and no pull-down	n/a	EXADC6
	PA7	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	EXADC7
	PC4	ADC1_IN14	Analog mode	No pull-up and no pull-down	n/a	VDADC14
	PC5	ADC1_IN15	Analog mode	No pull-up and no pull-down	n/a	VDADC15
	PB0	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	HSCADC8
	PB1	ADC1_IN9	Analog mode	No pull-up and no pull-down	n/a	HSCADC9
CAN1	PD0	CAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PD1	CAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
CAN2	PB13	CAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB5	CAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
DAC	PA4	DAC_OUT1	Analog mode	No pull-up and no pull-down	n/a	EXDAC4
	PA5	DAC_OUT2	Analog mode	No pull-up and no pull-down	n/a	EXDAC5
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
I2C3	PC9	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PA8	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
RCC	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-	RCC_OSC_OUT	n/a	n/a	n/a	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	OSC_OUT					
SPI2	PB10	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB12	SPI2_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI3	PA15	SPI3_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	
TIM1	PE8	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Low	HS1
	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	LS1
	PE10	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	HS2
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	LS2
	PE12	TIM1_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Low	RGB_B
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LS3
	PE14	TIM1_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	LS4
TIM2	PB11	TIM2_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	RGB_R
TIM3	PB4	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	RGB_G
TIM4	PB6	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
USART3	PD8	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PD9	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PD10	USART3_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
GPIO	PE2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP2
	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP3
	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP4
	PE5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP5
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP6
	PC13-ANTI_TAMP	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP7
	PB2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BOOT1
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Gelb
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROT
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP7
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP6
	PD13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP5
	PD14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP4
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP3
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP2
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP1
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX2GP0
	PD4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BTN_U
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP0
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EX1GP1

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1, ADC2 and ADC3 global interrupts	unused		
CAN1 TX interrupts	unused		
CAN1 RX0 interrupts	unused		
CAN1 RX1 interrupt	unused		
CAN1 SCE interrupt	unused		
TIM1 break interrupt and TIM9 global interrupt	unused		
TIM1 update interrupt and TIM10 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM11 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
SPI2 global interrupt	unused		
USART1 global interrupt	unused		
USART3 global interrupt	unused		
SPI3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	unused		
CAN2 TX interrupts	unused		
CAN2 RX0 interrupts	unused		
CAN2 RX1 interrupt	unused		
CAN2 SCE interrupt	unused		
USB On The Go FS global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
I2C3 event interrupt			unused
I2C3 error interrupt			unused
FPU global interrupt			unused

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware						
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing
DMA	ADC1 ✓	TIM1 ✓	CAN1 ✓			
GPIO ✓	DAC ✓	TIM2 ✓	CAN2 ✓			
NVIC ✓		TIM3 ✓	I2C1 ✓			
RCC ✓		TIM4 ✓	I2C3 ✓			
SYS ✓			SPI2 ✓			
			SPI3 ✓			
			USART1 ✓			
			USART3 ✓			
			USB_FS ✓			

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f405-415_407-417_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f405-415_407-417_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
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Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
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Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0182-stm32f405407xx-

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Datasheet	https://www.st.com/resource/en/datasheet/dm00037051.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0090-stm32f405415-stm32f407417-stm32f427437-and-stm32f429439-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn0516-overview-of-the-stm32f0xf100xxf103xx-and-stm32f2xxf30xf4xx-mcus-pmsm-singledual-foc-sdk-v40-stmicroelectronics.pdf
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